

Floating Point In TOY

16-bit format · 5-step algorithm using TOY integer operations

1. Extraction

Split the number into sign, exponent, mantissa; restore the implicit leading 1.

2. Alignment

Compare exponents; shift the smaller mantissa right until both are on the same scale.

3. Arithmetic

Same sign → add mantissas. Different sign → subtract smaller from bigger, keep bigger's sign.

4. Normalization

Too big → shift right, exponent +1. Too small → shift left, exponent -1.

5. Packing

Combine the final sign, exponent, and mantissa back into one 16-bit number.

```
// Extraction
unpack A -> signA, expA, mantA (add implicit bit)
unpack B -> signB, expB, mantB (add implicit bit)

// Alignment
diff = expA - expB
IF diff > 0: mantB >>= diff, resultExp = expA
ELSE:      mantA >>= -diff, resultExp = expB

// Arithmetic
IF signA == signB:
    resultMant = mantA + mantB
    resultSign = signA
ELSE:
    resultMant = |mantA - mantB|
    resultSign = sign of bigger mantissa

// Normalization
WHILE resultMant overflows: shift right, resultExp += 1
WHILE resultMant missing leading 1: shift left, resultExp -= 1

// Packing
RETURN combine(resultSign, resultExp, resultMant without implicit bit)
```

Pseudocode